

The Exceptional-Prime Desert Map

$$\alpha_0 = 299 + \pi/10 \quad \text{rule: } p \text{ exceptional} \iff r = p \|p\pi/10\| < 1$$

Key constants

Quantity	Value
$\kappa = \varphi c/108$ (15 digits)	4.84330141945946
Bost bound $2\sqrt{13}$	7.21110255
Single-axis budget	0.600925
Classical set S_4	{2, 3, 19, 191}
$C_{\text{unwt}}(S_4) = C_{\text{unwt}}(S_{20})$	1.433676813 ($< 2\sqrt{13}$)
$C_{\text{wt}}(S_4)$	11.42214869
$C_{\text{wt}}(S_5)$ (at the bridge)	40.43789948
$C_{\text{wt}}(S_{20})$	40793.25685
κ -axis share at P_5	2.534298×10^9
Number of exceptional primes $< 10^{4000}$	20

H4 classical invariants ($I_k = \log p_k/(p_k - 1)$)

The four real Weyl-axis shares of the 600-cell reading (other eight axes are zero).

Axis	Prime p	I_k
1	2	0.693147
2	3	0.549306
3	19	0.163580
4	191	0.0276435
Sum $\sum I_k$		1.43368 (< 7.211)

The bridge at P_5 (the one giant verifiable by hand)

$$P_5 = 3,993,746,143,633, \quad q = 191 (\in S_4), \quad m = 16, \quad k = 4302500806252,$$

$$|q\kappa^m - P_5 - k\pi| = 0.0382906 \pm 0.029 < 1, \quad \text{error ratio} = 9.588 \times 10^{-15}.$$

For P_1 – P_4 the relation is 0 exactly ($q = p$, $m = 0$, $k = 0$). For P_6 – P_{20} it is *beyond tolerance* (propagated κ -uncertainty $10^{1.9} \dots 10^{3537}$): marked **indeterminate**, never a pass.

Table 1 — The twenty exceptional primes

D_k = number of digits; w_k = desert width (run of consecutive non-exceptional integers ending just below p_k); $r_k = p_k \|p_k\pi/10\|$. Giants shown head...tail.

k	D_k	Prime p_k	Desert width w_k	r_k	Regime
1	1	2	0	0.7433629	Classical-Z
2	1	3	0	0.1725666	Classical-Z
3	2	19	15	0.5885052	Classical-Z
4	3	191	171	0.8441596	Classical-Z
5	13	3993746143633	3993746143441	0.1523629	Bridge
6	16	3224057731518397	3220063985374763	0.7749508	Desert
7	24	631474305334326148720631	631474302110268417202233	0.1441999	Desert

		k	D_k	Prime p_k	Desert width w_k	r_k	Regime		
8	35	10531012662744699702276055940873441			10531012662113225396941729792152809	0.3570549	Desert		
9	76	76365550274329238633...274704893143			76365550274329...64019701	0.4366733	Desert		
10	95	87822897782612089843...462065410669			87822897782612...60517525	0.8334696	Desert		
11	111	48983077336683228790...692784889819			48983077336683...19479149	0.5405288	Desert		
12	372	19462609931099711093...516236611721			19462609931099...51721901	0.5248112	Desert		
13	859	88321609940533775518...058013286951			88321609940533...76675229	0.5359789	Desert		
14	1025	19681903024283226243...591371595031			19681903024283...58308079	0.0147607	Desert		
15	1592	52943490324271006099...752677909607			52943490324271...06314575	0.9803959	Desert		
16	1863	17494378896143817560...864658557771			17494378896143...80648163	0.0504900	Desert		
17	2272	91511748796834962231...086060269167			91511748796834...01711395	0.5457596	Desert		
18	2389	20716339208351233251...013122521601			20716339208351...62252433	0.6931474	Desert		
19	3428	99469874438421417033...698029440779			99469874438421...06919177	0.5105710	Desert		
20	3548	12737470687711639675...504644341493			12737470687711...14900713	0.0276853	Desert		

Table 2 — Weighted energy and bridge status

C_{wt} is cumulative; H4 = active symmetry axes; $\log_{10} \tau$ is the κ -tolerance (digits of κ needed to test the bridge).

k	C_{wt} (cum.)	H4	BDP status	error	$\log_{10} \tau$
1	1.386294	12	EXACT	0	—
2	3.034213	12	EXACT	0	—
3	6.142232	12	EXACT	0	—
4	11.42215	12	EXACT	0	—
5	40.43790	1	VERIFIED	0.0382906	−1.54
6	76.14732	1	BEYOND_TOLERANCE	—	1.6
7	130.9497	1	BEYOND_TOLERANCE	—	10.0
8	209.2893	1	BEYOND_TOLERANCE	—	20.4
9	384.0161	1	BEYOND_TOLERANCE	—	62.6
10	602.6319	1	BEYOND_TOLERANCE	—	81.8
11	857.5051	1	BEYOND_TOLERANCE	—	97.6
12	1712.430	1	BEYOND_TOLERANCE	—	358.7
13	3690.226	1	BEYOND_TOLERANCE	—	846.7
14	6048.751	1	BEYOND_TOLERANCE	—	1012.2
15	9713.830	1	BEYOND_TOLERANCE	—	1579.8
16	14001.80	1	BEYOND_TOLERANCE	—	1850.4
17	19233.19	1	BEYOND_TOLERANCE	—	2260.2
18	24732.49	1	BEYOND_TOLERANCE	—	2376.6
19	32625.75	1	BEYOND_TOLERANCE	—	3416.4
20	40793.26	1	BEYOND_TOLERANCE	—	3535.5

Honesty note. Desert widths and weighted energies are computed; the “bridge” is a numerical observation verifiable only at P_5 (and trivially P_1 – P_4); the 12-curve H4 reading is an *interpretation*, not a theorem. The “error rate” (a measurement) and the “self-symmetry” (the $12 \rightarrow 1$ Coxeter reading) are distinct. No Hodge / BSD / mass-gap claim is made and no new mathematics is asserted. Certified data file `data/desert_map.csv` is byte-identical (sha256 `e0ea8b28...2291cbb`).